

Bicycle Lock with Tamper Detection and Remote Alert

ABSTRACT

A bicycle lock includes a locking shackle, an accelerometer mounted within the lock housing, a wireless communication module, and a control circuit. The control circuit monitors accelerometer output for vibration patterns consistent with tampering while the lock is in a locked state, and transmits an alert notification to a paired mobile device upon detecting such a pattern.

BACKGROUND

Conventional bicycle locks provide only physical resistance to theft and do not notify an owner when tampering occurs, delaying discovery of an attempted theft until the owner returns to the bicycle. Incorporating tamper detection directly into the lock allows an owner to respond while a theft attempt is still in progress.

SUMMARY

In one embodiment, an accelerometer within the lock housing continuously samples vibration while the shackle is in a locked position. A control circuit compares the sampled vibration pattern against a stored tamper signature and, upon a match exceeding a configured confidence threshold, activates a wireless communication module to transmit an alert to a paired mobile application along with the lock's last known location.

CLAIMS

1. A bicycle lock comprising: a locking shackle; an accelerometer disposed within a housing coupled to the shackle; a control circuit coupled to the accelerometer and configured to detect a tamper vibration pattern; and a wireless communication module configured to transmit an alert in response to detection of the tamper vibration pattern. 2. The lock of claim 1, wherein the control circuit is further configured to compare a sampled vibration pattern against a stored tamper signature. 3. The lock of claim 1, wherein the alert includes a last known location of the lock.